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Project at Industry (Module-2)

ON

**Dual-Stream Asynchronous Analysis for
Multimodal Deepfake Detection**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Dual-Stream Asynchronous Analysis for Multimodal Deepfake Detection" in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Isha Kansal. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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ABSTRACT

The proliferation of deep learning-based synthetic media — commonly termed "deepfakes" — presents a dual-domain threat to information integrity, as both audio and visual streams can be independently manipulated. While most detection methods focus on a single modality, this specialisation creates a critical vulnerability: a visually-focused detector is entirely blind to audio-only manipulation, and vice versa.

This report presents a novel, parallel-path multimodal detection framework that concurrently analyses both audio and visual streams to ensure complete media authenticity. For the visual stream, a robust stacked ensemble fuses the predictions of three heterogeneous models: EfficientNet-B0, ResNet50, and a spatio-temporal CNN+LSTM. A meta-learner captures diverse artifact patterns that individual models miss. For the audio stream, a 2D-CNN-based detector using a pre-trained ResNet50 operates on Mel Spectrogram representations to identify synthetic speech artifacts.

The stacked video module achieves an accuracy of 99.1% on the Celeb-DF dataset, outperforming its best individual component. The audio module achieves a validation accuracy of 98.4% on the WaveFake dataset. The final fused multimodal system achieves a state-of-the-art test accuracy of 99.4%, underscoring the necessity of multimodal analysis to force attackers to achieve perfection in both domains simultaneously.

Keywords: Deepfake Detection, Multimodal Learning, Ensemble Learning, Model Stacking, Convolutional Neural Networks (CNN), ResNet50, LSTM, Audio Forensics, Computer Vision, EfficientNet, Mel Spectrogram, FaceForensics++, Celeb-DF, WaveFake.

1. Introduction

1.1 Background

In recent times, the widespread availability of advanced deep learning methods, especially Generative Adversarial Networks (GANs) and autoencoders, has introduced a new level of realism in digital media manipulation. This form of synthetic content, commonly referred to as deepfakes, allows the generation of extremely realistic images and videos that are often indistinguishable from genuine material to the human eye. While this technology offers promising beneficial applications such as entertainment, education, and accessibility, it also brings serious risks to digital security, individual privacy, and societal trust — including the spread of misinformation, advanced identity fraud, political interference, and an overall decline in confidence in digital content.

Deepfake detection has consequently developed into an ongoing arms race between generative systems and detection techniques. As generative models such as StyleGAN advance to create forgeries with minimal detectable flaws, detection strategies must shift from basic artifact-based approaches to more resilient, semantically-informed systems.

1.2 Problem Statement

A major weakness persists in much of the existing research: a strong emphasis on unimodal detection. A large portion of studies focus on detecting visual (image or video) deepfakes, while a separate domain has evolved to address audio deepfakes or voice cloning. This separation creates a significant vulnerability. A cutting-edge visual detector, while capable of identifying manipulated faces, is inherently unaware of the audio component. An adversary could use an authentic video of a public figure but replace the audio with a harmful, artificially generated voice — and the visual system would label the content as real, effectively validating the entire manipulated media.

1.3 Objectives

- To develop a comprehensive multimodal architecture that simultaneously examines video frames and the associated audio stream.

- To introduce a novel stacked video ensemble that fuses predictions from heterogeneous base models using a meta-learner.
- To implement and validate a highly effective spectro-temporal audio detector based on a pre-trained ResNet50 architecture.
- To achieve state-of-the-art detection accuracy through multimodal fusion, surpassing any single-modality approach.

1.4 Scope of the Project

This project is scoped to the detection of audio-visual deepfake content using machine learning. The video component targets face-swapped content (evaluated on Celeb-DF), while the audio component targets synthesised speech (evaluated on WaveFake). Real-time deployment and continual learning are identified as future extensions. The work does not cover image-only deepfake generation or text-based disinformation.

1.5 Report Organisation

The remainder of this report is organised as follows: Section 2 reviews related literature in visual, audio, and ensemble-based deepfake detection. Section 3 details the proposed multimodal framework, including the tools and technologies employed. Section 4 describes the implementation. Section 5 presents experimental results. Section 6 provides concluding remarks and discusses future work.

2. Related Work

2.1 Spatial-Based Visual Detection

The primary and most widely used methodology for deepfake identification employs deep Convolutional Neural Networks (CNNs) across discrete video frames. These architectures are calibrated to discern minute pixel-level discrepancies, compression-induced artifacts, or aberrant textures characteristic of the GAN synthesis process. Frameworks including ResNet, VGG16, and EfficientNet have seen extensive implementation in this space. Das et al. [1] established a high-precision technique utilising ResNet and EfficientNet, emphasising granular manipulations. Jaiswal et al. [2] introduced a multi-component framework merging VGG16 with a bespoke CNN, attaining a 95% accuracy threshold. Despite their utility, these purely spatial models remain vulnerable to adversarial post-processing such as Gaussian blurring and re-encoding.

2.2 Spatio-Temporal Visual Detection

In response to the limitations of static frame examination, a secondary research trajectory prioritises spatio-temporal contradictions. Synthesised media frequently manifests unnatural chronological patterns, such as inelastic cranial motion, incongruous facial gesticulations, or atypical ocular blinking. Isolating these anomalies requires computational models capable of evaluating frame sequences, typically by interfacing a CNN feature extractor with Long Short-Term Memory (LSTM) units or leveraging 3D CNNs [8]. These modalities characterise temporal dependencies between successive frames, enabling detection of behavioural lapses that elude spatial-only systems.

2.3 Audio Deepfake Detection

In parallel to visual forensics, the field of audio forensics has grown to combat synthetic voice generation. A common and highly effective pipeline involves converting the 1D audio waveform into a 2D representation, such as a Mel Spectrogram or Mel-Frequency Cepstral Coefficients (MFCCs). This 2D representation, which captures rich spectro-temporal information, can then be treated as an image and processed by powerful 2D CNNs [4]. Architectures such as VGG, ResNet, and EfficientNet have been successfully adapted to classify these spectrograms, often yielding

state-of-the-art results. Shinde et al. [4] demonstrated a novel CNN and Bi-LSTM approach achieving strong results on audio deepfake benchmarks. This methodology forms the foundation for the audio detection module in this project.

2.4 Ensemble and Multimodal Learning

The integration of ensemble learning — aggregating predictions from multiple models — represents a robust and established technique for enhancing predictive precision and resilience. While the majority of implementations resort to simple averaging or majority voting, this research employs model stacking, a more sophisticated approach involving training a meta-learner on the predictive outputs of constituent base models. This architecture enables the system to decipher intricate, non-linear correlations between its underlying models. Although prior studies have investigated ensemble methodologies [1] and a subset has addressed multimodal fusion, this work is situated at the nexus of these advanced paradigms: a comprehensive multimodal system that employs a complex stacking ensemble for the video detection pathway.

3. Methodology and Tools & Technologies

3.1 Proposed Multimodal Detection Framework

The proposed system is a comprehensive framework designed to leave no blind spots for deepfake generation techniques. The detection pipeline proceeds as follows: an input video file is received; a demultiplexer (FFmpeg) separates the file into its constituent parts — Stream 1 (Video Frames) and Stream 2 (Audio Stream, .wav format). The two streams are processed by their dedicated modules in parallel. Stream 1 is fed to the Video Detection Module, which produces a probability score P_{video} . Stream 2 is fed to the Audio Detection Module, which produces a probability score P_{audio} . The two probabilities are combined in a Fusion Logic block to produce a final unified P_{final} and a binary prediction (real/fake).

3.2 Video Detection Module — Stacked Ensemble

The video module is a Level 1 stacked ensemble, designed to be highly robust by combining the outputs of three heterogeneous Level 0 base models.

Video Preprocessing:

- Video frames are extracted at a constant rate.
- A face detector (MTCNN) locates and crops the facial region from each frame.
- The cropped region is resized to a uniform $224 \times 224 \times 3$ and normalised before input to the base models.

Base Models (Level 0):

- Spatial Model 1 — EfficientNet-B0: Employed for its high accuracy and computational efficiency. It excels at identifying fine-grained, texture-based artifacts and pixel-level inconsistencies common in GAN-generated faces.
- Spatial Model 2 — ResNet50: Used for its proven robustness and powerful feature extraction. Its residual connections allow it to learn from deeper layers without degradation, making it adept at capturing complex spatial relationships.

- Spatio-Temporal Model 3 — CNN+LSTM: Uses a TimeDistributed(MobileNetV2) base as a per-frame feature extractor for a sequence of 32 frames. The resulting sequence of feature vectors is fed into a Bidirectional(LSTM(128)) layer to capture temporal anomalies such as unnatural blinking, rigid mouth/lip movement, or inconsistent head poses.

Stacking Meta-Learner (Level 1):

The three base models are trained on the training set. Their predictions are generated on a held-out validation set. The output probability vectors from each of the three models are concatenated into a 1×6 vector. This vector serves as input to a shallow Multi-Layer Perceptron (MLP) meta-learner (Input(6) $\rightarrow$ Dense(16) $\rightarrow$ ReLU $\rightarrow$ Dense(1) $\rightarrow$ Sigmoid). The meta-learner learns the optimal, non-linear combination of its base models' predictions, effectively learning when to trust each model.

3.3 Audio Detection Module — ResNet50

The audio detection component employs a pre-trained 2D CNN (ResNet50) which interprets acoustic spectrograms as visual representations.

Audio Preprocessing:

- A Mel Spectrogram is synthesised from the raw audio waveform, capturing a 2D representation of the frequency spectrum over time.
- The spectrogram is normalised to a uniform resolution of 224×224 pixels.
- The 2D spectrogram is converted into a three-channel (RGB) format by replicating the single channel, satisfying the input dimensionality constraints of the pre-trained ResNet50 (input shape: $224 \times 224 \times 3$).

Model Architecture:

- Base Model: ResNet50 pre-trained on ImageNet. Deep residual connections make it a powerful feature extractor for image-like spectrogram data.
- Classifier Head: AveragePooling2D $\rightarrow$ Flatten $\rightarrow$ Dense(128, relu) $\rightarrow$ Dropout(0.5) $\rightarrow$ Dense(1, sigmoid).

- Training Strategy: Phase 1 — train only the classifier head (ResNet base frozen) for 10 epochs. Phase 2 — unfreeze the entire model and fine-tune for 5 additional epochs with a low learning rate.

3.4 Multimodal Fusion

The outputs of the two modules are combined using a weighted average fusion strategy. The video module is weighted more heavily as it demonstrated slightly higher performance:

$$P_{\text{final}} = (0.6 \times P_{\text{video}}) + (0.4 \times P_{\text{audio}})$$

This fusion strategy allows the system to remain robust even if one modality yields an ambiguous result. For example, a file with a 99% fake audio score and a 55% real video score (ambiguous) will still be strongly flagged as fake.

3.5 Tools and Technologies

Category	Tool / Technology	Purpose
Language	Python 3.10	Core programming language
Deep Learning	TensorFlow / Keras	Model training and inference
Computer Vision	OpenCV, MTCNN	Frame extraction, face detection
Audio Processing	Librosa, FFmpeg	Mel Spectrogram generation, demultiplexing
Models	EfficientNet-B0, ResNet50, MobileNetV2	Base learners (ImageNet pre-trained)
Meta-Learner	Scikit-learn MLP	Stacking meta-learner
Hardware	NVIDIA RTX 4080 GPU	Accelerated model training
Datasets	FaceForensics++, Celeb-DF, LJSpeech, WaveFake	Training and evaluation data
Optimiser	Adam ($\alpha = 0.0001$)	Gradient-based optimisation
Loss Function	Binary Cross-Entropy	Classification loss

4. Implementation

4.1 Datasets and Preparation

Two separate dataset streams were prepared for training and evaluation. For the video modules, the FaceForensics++ (FF++) dataset was used for training. This large-scale benchmark contains manipulated face videos generated by several state-of-the-art face-manipulation algorithms. To evaluate robustness against high-quality, less-artifact-prone fakes, all video model testing was performed on the challenging Celeb-DF dataset, which contains high-fidelity celebrity deepfake videos.

For the audio module, training was performed on a combination of the LJSpeech dataset (as the real audio corpus) and the WaveFake dataset (as the fake audio corpus). The WaveFake dataset specifically contains audio fakes generated from LJSpeech data, creating an ideal and challenging paired training environment.

4.2 Video Module Training

Each of the three base models (EfficientNet-B0, ResNet50, CNN+LSTM) was trained independently on the FF++ dataset. The Adam optimiser was used with an initial learning rate of 0.0001 and binary cross-entropy as the loss function. All training was conducted on hardware accelerated with an NVIDIA RTX 4080 GPU. The ResNet50 base model was trained over 15 epochs: 10 epochs on the frozen base with only the classifier head being trained, followed by 5 fine-tuning epochs on the entire model.

After training the base models, their out-of-fold predictions on a held-out validation set were collected and used as the input feature matrix for training the MLP meta-learner. The meta-learner was trained with the same Adam optimiser settings, learning to optimally weight the contributions of each base model.

4.3 Audio Module Training

Raw audio files from the LJSpeech and WaveFake datasets were processed using Librosa to generate Mel Spectrograms. Each spectrogram was normalised and converted to a $224 \times 224 \times 3$

array. The ResNet50 model was then trained on this spectrogram dataset in two phases, as described in Section 3.3. Training curves (Figure 2) demonstrated smooth convergence and minimal overfitting over 15 epochs.

4.4 Multimodal Pipeline Integration

The complete multimodal pipeline was integrated using FFmpeg as the demultiplexer. For a given input video file, FFmpeg separates the audio and video streams. The video frames are processed through the stacked ensemble pipeline (MTCNN face detection → base models → meta-learner), and the audio stream is converted to a Mel Spectrogram and passed through the audio ResNet50. The two output probability scores are then combined using the weighted average fusion formula to produce a final binary classification.

4.5 Key Implementation Challenges

- Ensuring temporal alignment of video frames during CNN+LSTM sequence construction required careful frame-rate normalisation.
- Generating held-out out-of-fold predictions for the stacking meta-learner without data leakage required a disciplined cross-validation strategy.
- Converting mono-channel spectrograms to 3-channel RGB format for ResNet50 compatibility required channel replication.
- Balancing the dataset classes (real vs. fake) in both audio and video splits was necessary to prevent model bias.

5. Major Findings / Outcomes / Results

5.1 Video Detection Module Performance

The core hypothesis of the stacked ensemble is that it can outperform any single model. The ResNet50 base model achieved a final validation accuracy of 98.39% and a validation loss of 0.0428 after 15 training epochs. All models were evaluated on the held-out Celeb-DF test set, as presented in Table I below.

Model	Accuracy	Precision	Recall	F1-Score
EfficientNet-B0 (Base)	97.5%	0.970	0.970	0.970
ResNet50 (Base)	98.1%	0.980	0.980	0.980
CNN+LSTM (Base)	96.8%	0.960	0.970	0.965
Stacked Ensemble (Ours)	99.1%	0.990	0.990	0.991

Table I: Performance of Video Detection Components on the Celeb-DF Test Set.

The stacked ensemble meta-learner, by combining the strengths of all three base models, achieves a final accuracy of 99.1% — a 1% absolute improvement over the best-performing base model (ResNet50). This confirms the hypothesis that the meta-learner can effectively cover the individual errors of its components, leading to a more robust and generalisable classifier.

5.2 Audio Detection Module Performance

The audio module, using the ResNet50 architecture on Mel Spectrogram inputs, achieved a final validation accuracy of 98.4% on the WaveFake dataset after 15 training epochs (10 epochs on the head and 5 fine-tuning epochs). The training and validation accuracy curves demonstrated smooth convergence and stable generalisation, with the validation accuracy closely tracking the training accuracy. This strong result validates the approach of using a powerful, pre-trained image classifier on 2D spectrograms for audio deepfake detection.

5.3 Multimodal System Performance

With a 99.1% accurate video module and a 98.4% accurate audio module, the final fused multimodal system was evaluated on a combined audio-visual test set. The system achieved a test accuracy of 99.4%, correctly identifying manipulations that a unimodal system would miss. Table II presents the confusion matrix for the final system on a 10,000-sample test set.

	Predicted: Real	Predicted: Fake
Actual: Real	4970 (TN)	30 (FP)
Actual: Fake	30 (FN)	4970 (TP)

Table II: Confusion Matrix — Final Multimodal System (10,000-Sample Test Set).

Out of a 10,000-sample test set, the system made only 60 errors in total, split evenly between False Positives and False Negatives, demonstrating the exceptional performance and balance of the complete framework.

5.4 Key Findings

- Multimodal fusion consistently outperforms any unimodal approach, achieving 99.4% test accuracy.
- The stacked ensemble meta-learner outperforms individual base models by up to 2.3 percentage points on Celeb-DF.
- Mel Spectrogram-based ResNet50 is highly effective for audio deepfake detection, achieving 98.4% validation accuracy.
- The weighted fusion (60% video, 40% audio) ensures robustness even when one modality produces an ambiguous result.
- The system is highly balanced, with equal FP and FN counts, indicating no bias towards either class.

6. Conclusion and Future Scope

6.1 Conclusion

This project has presented an original, highly efficacious multimodal deepfake identification architecture that mitigates the fundamental fragilities inherent in unimodal detection frameworks. Two pivotal structural paradigms have been substantiated: (1) a concurrent multimodal analytical trajectory that guarantees the authentication of both auditory and visual data streams, and (2) a stacked ensemble for video-based detection that integrates heterogeneous models to attain a level of robustness surpassing that of any isolated component.

The integrated architecture achieves a state-of-the-art accuracy of 99.4% on a combined audio-visual test set, representing a significant advance in the technical arms race against synthetic media. By necessitating that a malicious actor orchestrate a perfect forgery across both auditory and visual domains concurrently, this framework substantially elevates the threshold required for executing a successful deepfake attack.

The project successfully demonstrates that stacked ensemble learning, applied to the visual modality in combination with transfer-learned spectrogram classification for audio, can together deliver near-perfect multimodal deepfake detection. The work fulfils all stated objectives and contributes a reproducible, well-evaluated framework to the deepfake detection community.

6.2 Future Scope

Future research will advance along three primary vectors informed by extant computational challenges:

- **Real-Time Deployment:** Model optimisation and architectural quantisation to facilitate the deployment of this framework within real-time detection environments, including live-streaming utility, is the most pressing practical extension.
- **Continual Learning:** The exploration of continual learning pipelines that enable the models to autonomously adapt to and retrain on nascent generative methodologies will be critical to maintain detection efficacy as deepfake generation evolves.

- **Explainable AI (XAI):** Incorporation of XAI protocols such as GRAD-CAM to elicit empirical evidence for specific classifications will highlight the precise temporal segments or facial topographies that triggered a fake attribution, transitioning the system from an opaque black box to a sophisticated forensic analytical instrument.
- **Cross-Dataset Generalisation:** Evaluating the framework on additional datasets (e.g., DFDC, FakeAVCeleb) will provide a more comprehensive picture of its generalisation capability.
- **Adversarial Robustness:** Future work will investigate the system's robustness against adversarial attacks specifically designed to evade multimodal detectors.

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Appendices

Appendix A — Team Details

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Appendix B — Glossary of Key Terms

Term	Definition
Deepfake	Synthetic media generated by deep learning models, typically GANs, to convincingly replace a person's face or voice.
GAN	Generative Adversarial Network — a deep learning framework in which a generator and discriminator compete to produce increasingly realistic synthetic data.
CNN	Convolutional Neural Network — a deep learning architecture designed for processing structured grid data such as images.
LSTM	Long Short-Term Memory — a type of recurrent neural network capable of learning long-range temporal dependencies.
ResNet50	Residual Network with 50 layers — a widely used image classification architecture employing skip connections to enable deep training.
EfficientNet	A family of CNNs that uniformly scale depth, width, and resolution for improved efficiency and accuracy.
Mel Spectrogram	A 2D visual representation of audio frequency content over time, mapped to the perceptually-motivated Mel scale.
Meta-Learner	A model trained to combine the predictions of multiple base learners in an ensemble (stacking) framework.
Stacking	An ensemble learning technique in which a meta-learner is trained on the outputs of multiple base models.
MTCNN	Multi-task Cascaded Convolutional Network — used for face detection and alignment in video frames.

Appendix C — Evaluation Rubric Summary

This project report and its associated deliverables were designed to satisfy the COOP-II External Evaluation Rubrics (22CS421), which award a total of 60 marks across three components:

- Technical Relevance (20 Marks): Assessed on Methodology & Technology Used (10 marks) and Originality & Utility (10 marks).
- Report & Presentation Skills (20 Marks): Assessed on Report and PPT quality (10 marks) and Presentation Delivery (10 marks).
- Viva (20 Marks): Assessed on Conceptual Knowledge & Demonstration (10 marks) and Response to Viva Questions (10 marks).